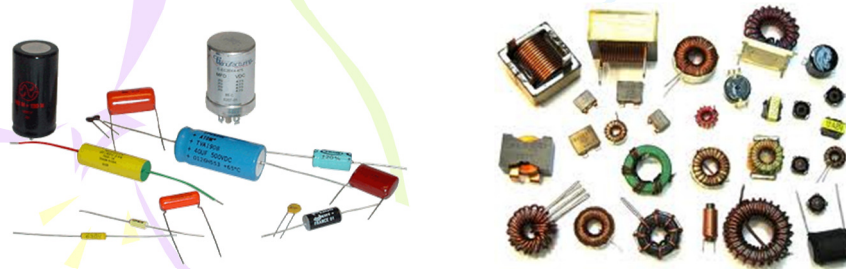


EBN 111/122
ELECTRICITY AND ELECTRONICS
Chapter 6
Capacitors and Inductors



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Capacitors and Inductors
Chapter 6

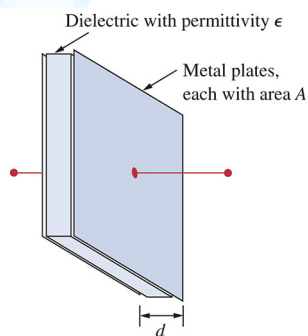
- 6.2 Capacitors
- 6.3 Series and Parallel Capacitors
- 6.4 Inductors
- 6.5 Series and Parallel Inductors

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6.2 Capacitors

Physical characteristics

- A capacitor is a passive element designed to **store energy** in its **electric field**.
- A **capacitor** consists of two conducting plates separated by an insulator (or dielectric).



$$C = \frac{\epsilon A}{d}$$

A = Surface area of the metal plate [m^2]

ϵ = permittivity of the dielectric [F/m]

d = distance between plates [m]

$\epsilon = \epsilon_r \epsilon_0$

ϵ_r = Relative permittivity of material

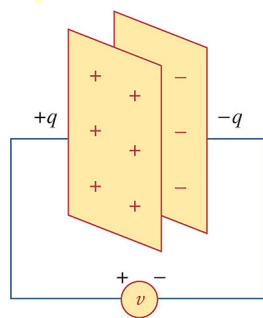
ϵ_0 = Permittivity of air ($8.854 \times 10^{-12} \text{ F/m}$)

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6.2 Capacitors

Electrical characteristics

- Capacitance** C is the ratio of the charge q on one plate of a capacitor to the voltage difference v between the two plates, measured in farads (F).



$$C = \frac{q}{v}$$

$$q = Cv$$

$$\frac{dq}{dt} = \frac{d}{dt}(Cv) = C \frac{dv}{dt}$$

$$i = C \frac{dv}{dt}$$

If $i > 0$: Capacitor is charging.
If $i < 0$: Capacitor is discharging.

OR

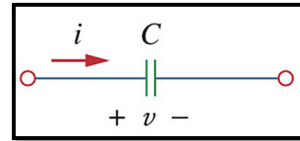
$$v = \frac{1}{C} \int_{-\infty}^t i dt = \frac{1}{C} \int_{t_0}^t i dt + v(t_0)$$

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6.2 Capacitors

Energy and Power

- The energy, **w**, stored and power **p** in the capacitor is:



$$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$$

$$p = vi = C v \frac{dv}{dt}$$

- Important characteristics of capacitors:

- It is an **open circuit** to dc ($dv/dt = 0$).
- Its voltage **cannot change abruptly** ($dv/dt = \infty$).
- Does not dissipate energy, but stores it when charging or provides it when discharging.

$$i = C \frac{dv}{dt}$$

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6.2 Capacitors

New Equations

$$q = C v$$

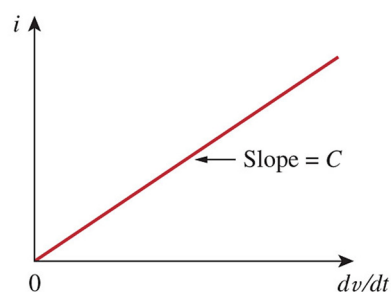
$$C = \frac{\epsilon A}{d}$$

$$i = C \frac{dv}{dt}$$

$$v = \frac{1}{C} \int_{t_0}^t i dt + v(t_0)$$

$$w = \frac{1}{2} C v^2$$

V-I characteristic of a Cap



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PP 6.1

What is the voltage across a $3\mu\text{F}$ capacitor if the charge on one plate is 0.12 mC . How much energy is stored in the capacitor?

$$v = \frac{q}{C} =$$

$$w = \frac{1}{2} C v^2 =$$

PP 6.2

The voltage source connected across a $10\mu\text{F}$ capacitor is given as:
 $v(t) = 50 \sin 2000t\text{ V}$. Calculate $i(t)$.

$$i = C \frac{dv}{dt} =$$

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PP 6.3

If the current through a $100\mu\text{F}$ capacitor is given as:
 $i(t) = 50 \sin(120\pi t)\text{ mA}$, calculate $v(1\text{ms})$ and $v(5\text{ms})$.
 Assume $v(0) = 0\text{ V}$

$$v = \frac{1}{C} \int_{t_0}^t i\,dt + v(t_0)$$

$$v(1\text{ms}) = \frac{1}{C} \int_0^{1\text{ms}} 50 \sin(120\pi t) \times 10^{-3}\,dt + 0$$

$$=$$

$$=$$

$$=$$

$$v(5\text{ms}) = \frac{1}{C} \int_0^{5\text{ms}} 50 \sin(120\pi t) \times 10^{-3}\,dt + 0 = 1.736\text{ V}$$

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PP 6.5

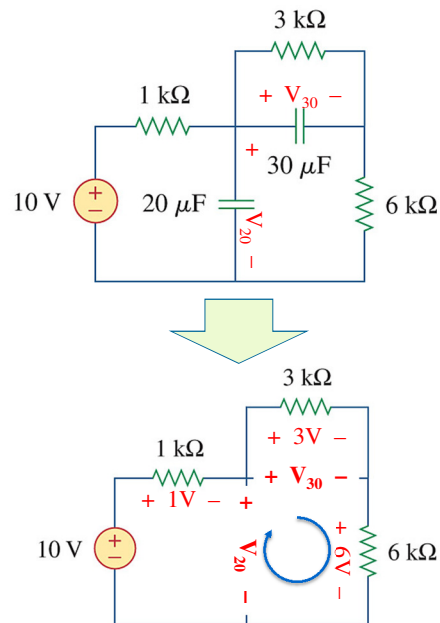
For DC conditions, calculate the energy stored in the two capacitors.

$$V_{30\mu\text{F}} = 3 \text{ V}$$

$$\begin{aligned} w &= 0.5 C v^2 \\ &= 0.5 \times 30 \times 10^{-6} \times 3^2 \\ &= 135 \mu\text{J} \end{aligned}$$

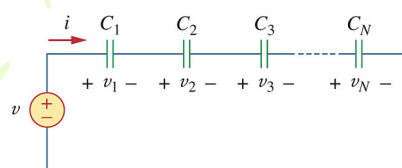
$$V_{20\mu\text{F}} =$$

$$\begin{aligned} w &= \\ &= \\ &= \end{aligned}$$

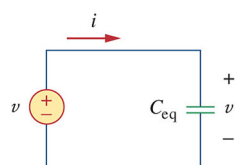


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6.3 Series and Parallel Capacitors

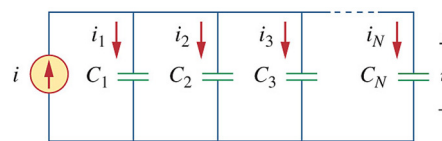


(a)

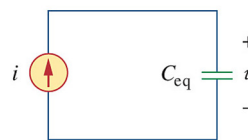


(b)

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$



(a)



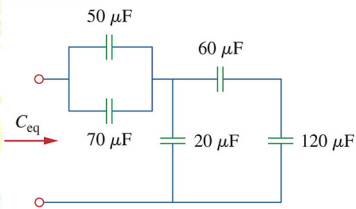
(b)

$$C_{eq} = C_1 + C_2 + \dots + C_N$$

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PP 6.6

Determine the equivalent capacitance



$$(C_{eq} = 40\ \mu\text{F})$$

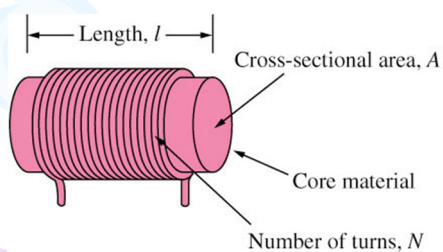
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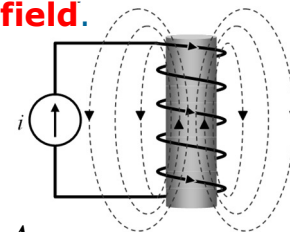
6.4 Inductors

Physical characteristics

- An inductor is a passive element designed to **store energy** in its **magnetic field**.



$$L = \frac{N^2 \mu A}{l} [H]$$



- An inductor consists of a coil of conducting wire.
- Core from iron, steel, plastic or air.
- μ = permeability = $\mu_r \mu_0$

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6.4 Inductors

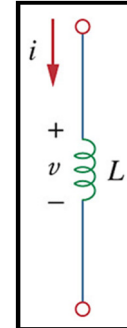
Electrical characteristics

$$v = L \frac{di}{dt}$$

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

$$w = \frac{1}{2} Li^2$$

$$p = vi = L \frac{di}{dt} i$$



- Important characteristics of inductors:
 - It is a **short circuit** to DC ($di/dt = 0$).
 - Its **current** cannot change abruptly ($di/dt = \infty$).
 - Does not dissipate energy, but stores it when charging or provides it when discharging.

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PP 6.8

The current through a 1 mH inductor is $i(t) = 20 \cos 100t$ mA. Calculate the voltage and the stored energy.

$$v = L \frac{di}{dt} = 1 \times 10^{-3} \times \frac{d}{dt} \{20 \cos 100t \times 10^{-3}\}$$

=

=

$$w = \frac{1}{2} Li^2 = 0.2 \cos^2 100t \mu J$$

PP 6.9

The voltage across a 2 H inductor is $v(t) = 10(1 - t)$ V. Determine the current through the inductor at $t = 4$ s if $i(0) = 2$ A.

$$i(4) = \frac{1}{2} \int_0^4 10(1 - t) dt + 2$$

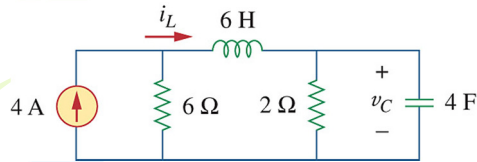
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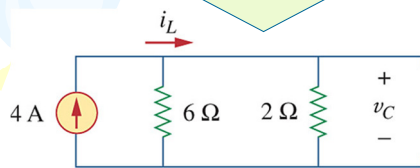
PP 6.10

Determine v_C , i_L and the energy stored in both the capacitor and the inductor for the DC circuit below.



$$i_L = 4 \frac{6}{6 + 2} = 3 \text{ A}$$

$$w = \frac{1}{2} L i^2 = 0.5 \times 6 \times 3^2 = 27 \text{ J}$$

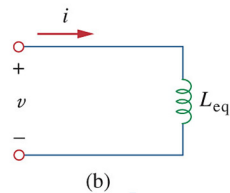
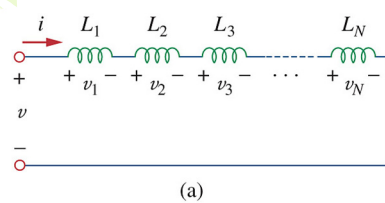


$$v_C =$$

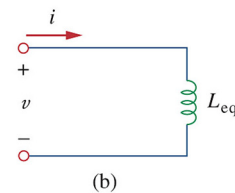
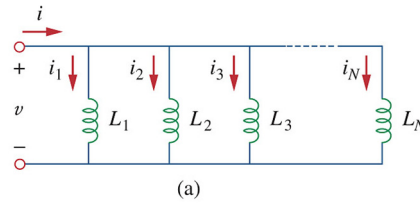
$$w =$$

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6.4 Series and Parallel Inductors



$$L_{eq} = L_1 + L_2 + \dots + L_N$$



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$$

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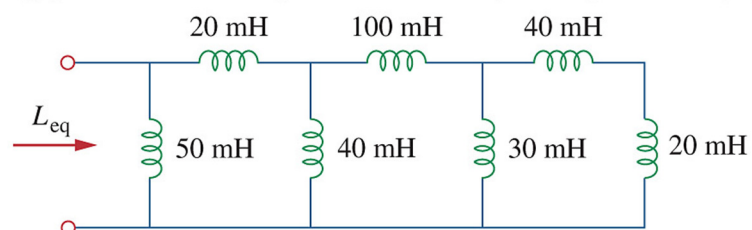
Current and voltage relationship for R, L, C

Circuit element	Units	Voltage	Current	Power
 Resistance	ohms (Ω)	$v = Ri$ (Ohm's law)	$i = \frac{v}{R}$	$p = vi = i^2 R$
 Inductance	henries (H)	$v = L \frac{di}{dt}$	$i = \frac{1}{L} \int v dt + k_1$	$p = vi = Li \frac{di}{dt}$
 Capacitance	farads (F)	$v = \frac{1}{C} \int i dt + k_2$	$i = C \frac{dv}{dt}$	$p = vi = Cv \frac{dv}{dt}$

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PP 6.11

Calculate L_{eq}



$$L_{eq} = 25 \text{ mH}$$

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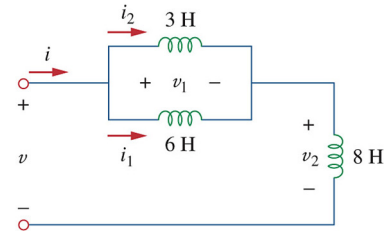
PP 6.12

If $i_1(t) = 0.6e^{-2t}$ A and $i(0) = 1.4$ A, find

- a) $\underline{i_2(0)}$ (b) $i_2(t)$ and $i(t)$ (c) $v_1(t)$, $v_2(t)$ and $v(t)$

a) $i_1(0) = 0.6$ A

$$\text{KCL: } i_2(0) = i(0) - i_1(0) \\ = 1.4 - 0.6 = 0.8 \text{ A}$$



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PP 6.12 continued

If $i_1(t) = 0.6e^{-2t}$ A and $i(0) = 1.4$ A, find

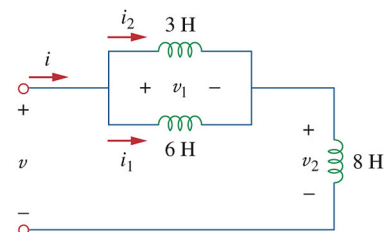
- (b) $\underline{i_2(t)}$ and $i(t)$ (c) $\underline{v_1(t)}$, $v_2(t)$ and $v(t)$

a) $i_2(0) = 0.8$ A

$$i_2(t) = \frac{1}{L} \int v_1 dt + i_2(0) \quad \therefore \text{need } v_1$$

$$v_1(t) = L_1 \frac{di_1}{dt} = 6 \frac{d}{dt} \{0.6e^{-2t}\} \\ = 6 \times -2 \times 0.6e^{-2t} \\ = -7.2e^{-2t} \text{ V}$$

$$i_2(t) = \frac{1}{3} \int_0^t -7.2e^{-2t} dt + 0.8 = \frac{-7.2}{3} \left[\frac{1}{-2} e^{-2t} \right]_0^t + 0.8 \\ = 1.2[e^{-2t} - 1] + 0.8 \\ = 1.2e^{-2t} - 0.4 \text{ A}$$



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PP 6.12 continued ...

If $i_1(t) = 0.6e^{-2t}$ A and $i(0) = 1.4$ A, find

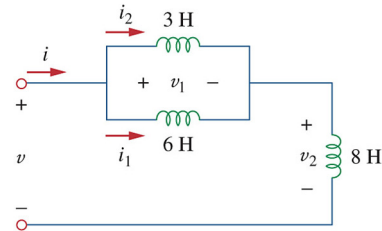
(b) $i_2(t)$ and $\underline{i(t)}$ (c) $v_1(t)$, $\underline{v_2(t)}$ and $\underline{v(t)}$

$$v_1(t) = -7.2e^{-2t} \text{ V}$$

$$i_2(t) = 1.2e^{-2t} - 0.4 \text{ A}$$

KCL:

$$\begin{aligned} i(t) &= i_1(t) + i_2(t) \\ &= 0.6e^{-2t} + 1.2e^{-2t} - 0.4 \text{ A} \\ &= \underline{1.8e^{-2t} - 0.4 \text{ A}} \end{aligned}$$



$$\begin{aligned} v_2(t) &= L \frac{di}{dt} = 8 \frac{d}{dt} \{1.8e^{-2t} - 0.4\} \\ &= 8 \times -2 \times 1.8e^{-2t} \\ &= \underline{-28.8e^{-2t} \text{ V}} \end{aligned}$$

KVL:

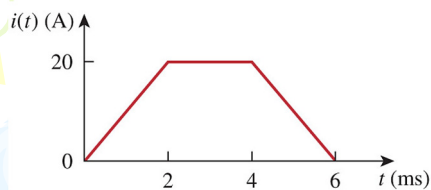
$$\begin{aligned} v(t) &= v_1(t) + v_2(t) \\ &= -7.2e^{-2t} - 28.8e^{-2t} \\ &= \underline{-36e^{-2t} \text{ V}} \end{aligned}$$

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Problem 6.40

The current through a 10 mH inductor is shown below.

Determine the voltage across the inductor at $t = 1, 3$ and 5 ms.



$$v(t) = 0.01 \frac{di}{dt} = 0.01 \frac{\Delta i}{\Delta t}$$

$$v(1\text{ms}) =$$

$$v(3\text{ms}) =$$

OR

$$i(t) = \frac{\Delta i}{\Delta t} t + c = \frac{20 - 0}{0.002 - 0} t + 0 =$$

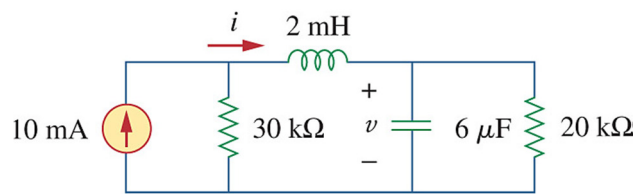
$$v(t) = 0.01 \frac{di}{dt} =$$

$$v(5\text{ms}) =$$

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Problem 6.48

Under steady state
dc conditions, find
 i and v .



$i =$

$v =$

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